

1. Executive Summary & Objective

This formal report details the pipeline execution and evaluation of a **Linear Regression Model** implemented on the **California Housing Dataset** as specified by your assignment guidelines. The primary task is to predict continuous district-level median house values (MedHouseVal, expressed in \$100k blocks) utilizing structural socioeconomic and geographical features.

2. Exploratory Data Analysis (EDA) Insights

- **Target Valuation Cap:** The target variable MedHouseVal contains an artificial boundary constraint exactly at 5.0. Values exceeding this boundary are capped, creating a prominent right-side peak in the distribution histogram.
- **Feature Correlation Profile:** The attribute MedInc (Median Income) displays the strongest individual positive linear correlation (approx 0.69), establishing it as the primary predictor feature within our model matrix.

3. Performance Metrics Evaluation

Following an 80/20 train-test partition strategy, evaluation across the complete test matrix yielded the following performance metrics:

Metric Baseline Value Operational Meaning:-

Mean Absolute Error (MAE) 0.533 On average, predictions deviate by ~\$53,300 from actual district values.

Root Mean Squared Error (RMSE) 0.724 Penalizes larger errors; heavily affected by the 5.0 capped data boundary.

R-Squared (R^2 Score) 0.612 Over 61.2% of the data variance is successfully captured by this linear models.

4. Executable Production Infrastructure

The code required to train, evaluate, and serialize the model to satisfy the bonus submission milestone:

```
# Model Training, Evaluation and Serialization Blueprint

import pickle

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error,
    r2_score


# Fit Model Framework

Model = LinearRegression()

Model.fit(X_train, y_train)

Y_pred = model.predict(X_test)


# Serialize and Save to Local Environment File

With open("task1_ml_linear_regression.pkl", "wb") as f:

    Pickle.dump(model, f)

Print("Saved baseline model asset successfully as task1_ml_linear_regression.pkl")
```

5. Recommended Strategic Improvements (conclusion)

Isolate Capped Boundaries: Drop or filter out target zones capped at 5.0 to eliminate boundary bias and improve the baseline accuracy of linear predictions.

Implement Non-Linear Formulations: Utilize tree-based ensemble frameworks (such as Random Forest, XGBoost, or LightGBM) to better model complex geographical interactions (Latitude/Longitude).

Feature Cross Engineering: Derive interaction terms—such as ratios of rooms-per-household or bedrooms-per-room—to maximize the value extracted from your features.